Double-Ended Queue (Deque) Introduction

Q. What is a Double-Ended Queue (Deque)? Explain its types and how it differs from a standard queue and stack. Provide real-world applications.

> **Definition to Remember**

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1. Operations in a Deque

Unlike standard queues (insert Rear, delete Front) or stacks (insert Top, delete Top), a Deque supports **four primary operations**:

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Visual Representation:

```text

## 2. Types of Restricted Deques

To enforce specific behaviors, Deques are classified into two restricted variants:

| Type | Restriction | Allowed Operations |

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## 3. Comparison with Stack and Queue

| Data Structure | Insertion Point | Deletion Point | Principle |

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## 4. Real-World Applications

- **Web Browser History:** When navigating back and forth, URLs are added/removed from one end. If history exceeds the max limit, oldest URLs are dropped from the other end.

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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